

Wednesday warm-up: displacement, velocity, and acceleration vectors

Prof. Jordan C. Hanson

August 27, 2025

1 Memory Bank

1. $\Delta x = \vec{x}_f - \vec{x}_i$... Definition of displacement
2. $\Delta t = t_f - t_i$... Definition of time duration
3. $\vec{v} = \Delta \vec{x} / \Delta t$... Definition of vector velocity
4. $\vec{a} = \Delta \vec{v} / \Delta t$... Definition of vector acceleration

2 Displacement, Velocity, and Time

1. Suppose the location of an aircraft is described by a 2D coordinate system in which East corresponds to the positive x-axis, and North corresponds to the positive y-axis, and altitude corresponds to the z-axis. The plane takes off from the origin in a direction 45 degrees above the xy-plane heading North, with a speed of 200 km/hr.

(a) Determine the components of the velocity, and write it as a vector $\vec{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k}$.

(b) What is the plane's location after 6 minutes?

(c) Five minutes into the flight, the plane's angle above the xy-plane changes to 0 degrees, and it also turns due East. The *speed* remains 200 km/hr for another 6 minutes. What is the final location?

the right and slowing down. (3) The vehicle is moving to the left and speeding up. (4) The vehicle is moving to the left and slowing down.

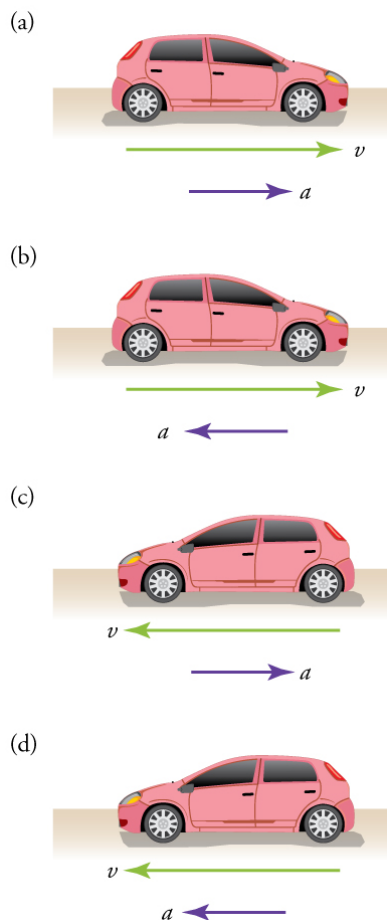


Figure 1: Four cases of velocity and acceleration in one dimension.

2. Suppose the velocity of the vehicle in Fig. 1 is 40 km hr⁻¹ at $t = 0$ seconds, and 80 km hr⁻¹ at $t = 5$ seconds. (a) What is the acceleration in km hr⁻¹ s⁻¹? (b) What is the acceleration in m s⁻²?

3 Acceleration

1. A vehicle with different velocity and *acceleration* vectors is shown in Fig. 1. Match the pictures to the following statements. (1) The vehicle is moving to the right and speeding up. (2) The vehicle is moving to